Dr. Rabia Khan

AI Research Scientist | Recommendation Systems | Applied Machine Learning
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PROFESSIONAL SUMMARY

PhD-qualified Artificial Intelligence researcher and Assistant Professor at the National University of Sciences and Technology (NUST), Pakistan, specializing in recommendation systems, graph neural networks, deep learning and multimodal AI. Research experience in cross-domain recommendation, data sparsity, cold-start problems, contrastive representation learning and graph-based modelling.
Hands-on research experience with Python, PyTorch, transformer architectures, parameter-efficient fine-tuning and vision-language models. Developed research solutions for cross-domain personalization and Urdu Nastaliq OCR using large vision-language models. Published in peer-reviewed international journals and conferences.
Interested in developing intelligent, scalable and personalized AI systems, particularly in recommendation, content understanding, multimodal learning and trustworthy AI.

RESEARCH INTERESTS

- Recommendation Systems
- Large-Scale Personalization
- Graph Machine Learning
- Applied AI
- Generative AI
- Trustworthy AI

PROFESSIONAL EXPERIENCE

Senior Data Analyst | Intelligence Unit, Quetta *Aug 2014 - Aug 2018*

- Established a data analysis facility supporting real-time situation monitoring, information analysis, and operational decision-making.
- Led the collection, classification, validation, analysis, and dissemination of multi-source information for development projects and operational monitoring.
- Developed dashboards, statistical visualizations, analytical reports, and performance-monitoring outputs using Power BI, Tableau, and Python.
- Conducted HR data analysis and developed KPIs and performance indicators to support organizational performance monitoring.
- Acquired, cleaned, compiled, validated, and analyzed data from disaster- and calamity-affected areas.
- Integrated information from multiple sources to support situation analysis, reporting, and operational planning.
- Coordinated information sharing and data exchange with NGOs and INGOs involved in development and disaster-response activities.
- Supported the transformation of raw operational information into structured datasets, analytical outputs, and decision-support information.

Research Associate | AI Center, NUST *Aug 2023 - Aug 2024*

- Directed a cross-functional team to improve recommendation systems through metadata-based modelling
- Developed AI-driven data pipelines for information extraction and semantic inference from unstructured data
- Led research in graph-based data modelling and large-scale data quality assessment

Lecturer, Assistant Professor & Senior Lecturer (progressive roles) | NUST, Software Engineering Dept.
2009- 2014 & 2024 - Present

- Contribute to academic programme development, quality assurance, assessment, and data-informed education processes within the Software Engineering programme.
- Teach and supervise graduate research in AI, machine learning, and data-driven systems.

- Guide applied projects in data integration, graph-based modelling, and generative AI systems.
- Supervise student research applying AI methods to real-world, cross-domain datasets.

SELECTED DATA & INFORMATION SYSTEMS PROJECTS

Cross-Domain Recommendation Using Graph Learning and Contrastive Representation Learning | 2018-2025 (Paper published at Journal of Supercomputing- 2025)

- Investigated recommendation challenges arising from data sparsity and cold-start conditions.
- Developed graph-based approaches to model user-item relationships across domains.
- Built a contrastive, attention-based framework (BERT behaviour encoder, contrastive self-supervised autoencoder, multi-head attention fusion, RoBERTa sentiment features) that transfers preferences from a rich source domain to sparse target domains (Journal of Supercomputing, 2026).
- Evaluated on Amazon Movies→Books (2,028 target users) and Movies→Music (718 target users) with leave-one-out testing and 99 sampled negatives, against seven published baselines.
- Hit Rate@10: 0.543 (Books) and 0.542 (Music), versus 0.376 and 0.436 for the strongest baseline (+44% and +24%). Hit Rate@50: 0.602 and 0.711.
- Versus GA-MTCDR: HR@50 +56.8% and NDCG@50 +11.6% on Movies→Books; HR@50 +39.1% and NDCG@50 +9.7% on Movies→Music; statistically significant ($p < 0.05$, paired t-test).
- Ablation: contrastive learning raised HR@10 from 0.391 to 0.454 (Books) and 0.388 to 0.443 (Music); the full model reached 0.547 and 0.551, a gain of 40% and 42% over the domain-specific baseline.

CaseFlow - Protection Monitoring & Referral Data Pipeline (2026)

- Built an end-to-end digital information management pipeline covering KOBO XLSForm data collection, automated Python/SQL ETL, data quality assurance, and Power BI-ready reporting.
- Designed and validated a KOBO XLSForm for structured monitoring and referral data collection, incorporating vulnerability screening, severity classification, consent, and referral tracking.
- Developed a Python/SQL ETL workflow to deduplicate records, validate required fields, normalize inconsistent data, and maintain a data-quality audit trail.
- Designed a layered SQL data model comprising raw, normalized, cleaned, and quality-assurance datasets to support reliable reporting.
- Developed analytical queries and dashboards for caseload, severity, referral status, and data-quality monitoring.

Cross-Disciplinary Data Integration Platform (2023)

- Designed and implemented a cloud-based ETL pipeline to unify metadata from educational, environmental, and healthcare data sources
- Consolidated multi-source data into a Neo4j graph database to support structured, query-ready data access
- Enabled reliable, cross-institutional data access for downstream analysis and reporting

EDUCATION

Ph.D. Computer Science | National University of Sciences and Technology (NUST), Pakistan | 2025

Thesis: Cross-Domain Recommendation using Deep Learning. Research focused on integrating heterogeneous datasets using graph-based and machine-learning approaches for data-driven decision-making.

MS, Computer Science | NUST, Pakistan | 2009

Thesis: Developing a Watermarking Scheme for Audio-in-Image using Wavelet Transform

BE, Software Engineering | NUST, Pakistan | 2006

Final Degree Project: AODV Stable Route Discovery using Route Fragility Coefficient as a Metric

LANGUAGES

- Urdu - (Fluent oral and written)
- English - Fluent (oral and written)
- Punjabi - Fluent (oral and written)

SELECTED PUBLICATIONS

- Khan, R., Iltaf, N., Latif, R. et al. Cross-domain recommendation framework for enhanced personalization through contrastive learning. Journal of Supercomputing, Springer 82, 55 (2026). https://doi.org/10.1007/s11227-025-08187-6, January 2026
- Khan, R., Iltaf, N. "Enhancing Cross-Domain Recommendations through User Convergence and Contrastive Learning", 28th IEEE International Conference on Advanced Communications Technology (ICACT 2026), Korea, January 2026
- Khan, R., Iltaf, N. et al., "Alleviating Cold-Start User Challenges through Semantic Aware Multi-Aspect Graph Convolutional Networks", under second Revision, at User Modeling and User-Adapted Interaction
- Khan, R., Iltaf, N., Latif, R. et al., "Cross-Domain Recommendation based on Metadata using Graph Convolution Networks" in IEEE Access, vol. 11, pp. 90724-90738, August 2023, doi: 10.1109/ACCESS.2023.3307015
- Khan, R., Iltaf, N., Rehman, F. et al., "Metadata Based Cross-Domain Recommender Framework using Neighbourhood Mapping", in IEEE Asia Pacific Conference on Computer Science and Data Engineering (2023), Fiji, 4th-6th December 2023.